Exercise 1: Setting Up Junit

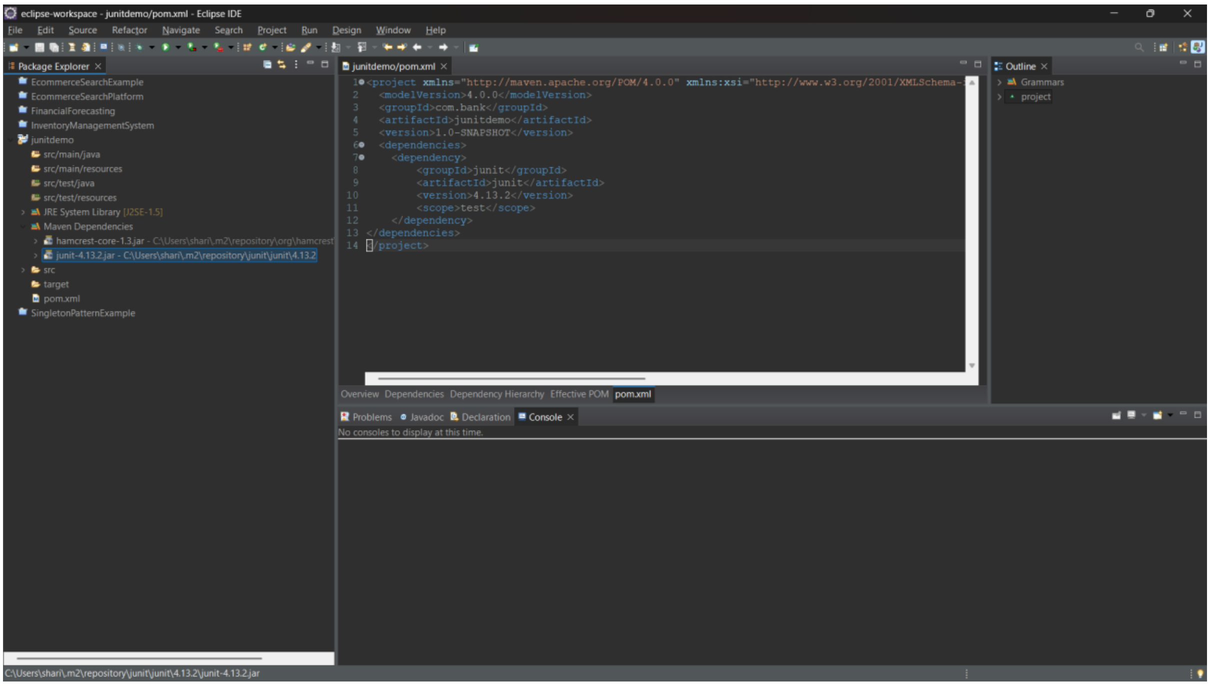
JUnit

- JUnit is a Java testing framework used for unit testing.
- It helps developers test individual units of code automatically.
- JUnit improves code quality and simplifies debugging.

Dependency Used

```
<dependency>
   <groupId>junit</groupId>
   <artifactId>junit</artifactId>
   <version>4.13.2</version>
   <scope>test</scope>
</dependency>
```

OUTPUT

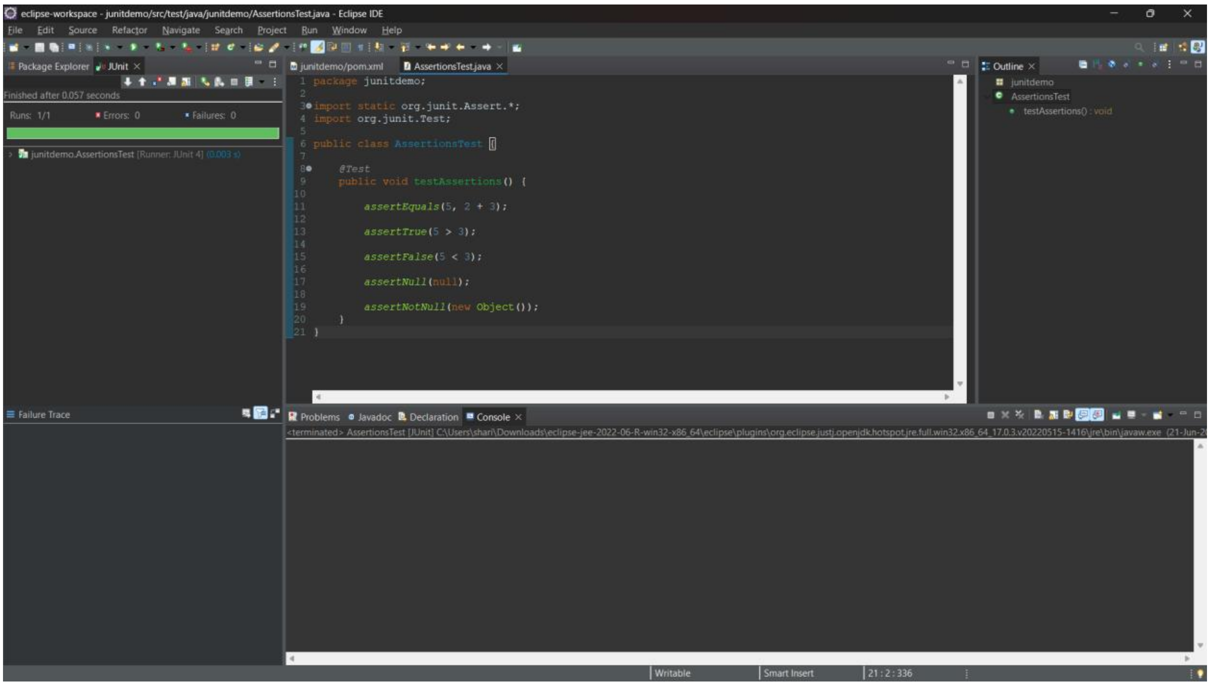


Exercise 3: Assertions in Junit

AssertionsTest.java

```
package junitdemo;
import static org.junit.Assert.*;
import org.junit.Test;
public class AssertionsTest {
  @Test
  public void testAssertions() {
    assertEquals(5, 2 + 3);
    assertTrue(5 > 3);
    assertFalse(5 < 3);
    assertNull(null);
    assertNotNull(new Object());
  }
}
```

OUTPUT



Exercise 4: (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in Junit

Arrange: Prepare the test data and required objects before executing the test.

Act: Execute the method or operation that needs to be tested.

Assert: Verify that the actual result matches the expected result using JUnit assertions.

@Before (Setup Method): The @Before annotation is used to initialize test data and resources before each test case is executed.

@After (Teardown Method): The @After annotation is used to clean up resources and reset the test environment after each test case execution.

Result: The test case executed successfully using the AAA pattern, with setup and teardown methods running before and after the test, resulting in **Runs: 1, Failures: 0, Errors: 0**.

OUTPUT